

QED Tree Amplitudes

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QED Feynman Rules

First, let us try to read out the Feynman rules from the Lagrangian. There are some subtlety, that we will discuss later

First, recall that the photon propagator is given by

$$\begin{array}{c} \xrightarrow{p} \\ \text{~~~~~} \end{array} = \frac{-i}{p^2 + i\epsilon} \left[g_{\mu\nu} - (1 - \xi) \frac{p_\mu p_\nu}{p^2} \right]. \quad (1.1)$$

For most cases, we will use the Feynman Gauge, $\xi = 1$, unless we are checking the gauge invariance explicitly. The photon propagator under the Feynman gauge is

$$\begin{array}{c} \xrightarrow{p} \\ \text{~~~~~} \end{array} = \frac{-ig_{\mu\nu}}{p^2 + i\epsilon} \quad (\text{Feynman gauge}). \quad (1.2)$$

A spinor propagator is represented by a solid line with an arrow pointing in the direction of particle flow (to the right for particles, left for antiparticles):

$$\begin{array}{c} \xrightarrow{p} \\ \text{—————} \end{array} = \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon}. \quad (1.3)$$

Just as in scalar QED, external photon lines get polarization vectors:

$$\begin{array}{c} \text{~~~~~} \end{array} \text{ (incoming)} = \epsilon_\mu(p) \quad (1.4)$$

$$\text{ (outgoing)} \text{~~~~~} = \epsilon_\mu^*(p) \quad (1.5)$$

Similarly, external fermions get spinors, with u spinors for electrons and v spinors for positrons:

$$\text{—————} \text{ (incoming)} = u^s(p), \quad (1.6)$$

$$\text{ (outgoing)} \text{—————} = \bar{u}^s(p), \quad (1.7)$$

$$\text{---} \leftarrow \text{---} \bigcirc = \bar{v}^s(p), \quad (1.8)$$

$$\bigcirc \text{---} \leftarrow \text{---} = v^s(p). \quad (1.9)$$

The external spinors are forced to be on-shell by the LSZ reduction formula. Thus, we can use the equations of motion to simplify the calculations:

$$(\not{p} - m)u^s(p) = \bar{u}^s(p)(\not{p} - m) = 0, \quad (1.10)$$

$$(\not{p} + m)v^s(p) = \bar{v}^s(p)(\not{p} + m) = 0. \quad (1.11)$$

Now, we shall investigate the interactions for QED. By expanding the Lagrangian, we find

$$\mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu A_\mu\psi. \quad (1.12)$$

Note, unlike scalar QED, there is no derivative, and thus no factor of momentum.

$$\text{---} \rightarrow \text{---} \begin{array}{l} \nearrow \text{---} \\ \searrow \text{---} \end{array} = -ie\gamma^\mu, \quad (1.13)$$

$$\text{---} \leftarrow \text{---} \begin{array}{l} \nearrow \text{---} \\ \searrow \text{---} \end{array} = -ie\gamma^\mu, \quad (1.14)$$

$$\begin{array}{l} \nearrow \text{---} \\ \searrow \text{---} \end{array} \text{---} = -ie\gamma^\mu, \quad (1.15)$$

$$\text{---} \begin{array}{l} \nearrow \text{---} \\ \searrow \text{---} \end{array} = -ie\gamma^\mu. \quad (1.16)$$

The μ on the γ^μ will be contracted with the μ of the photon, which would either be the ϵ_μ of a polarization vector or the $g_{\mu\nu}$ of the photon propagator.

If there is an internal fermion line between ends, the fermion propagator should go between the end spinors:

$$\begin{array}{c}
p_1 \quad \gamma^\nu \quad p_2 \quad \gamma^\mu \quad p_3 \\
\downarrow \quad \downarrow \\
q_1 \quad q_2 \\
\downarrow \quad \downarrow \\
\epsilon_1^\nu \quad \epsilon_2^\mu
\end{array}
= (-ie)^2 \bar{u}(p_3) \gamma^\mu \frac{i(\not{p}_2 + m)}{p_2^2 - m^2 + i\epsilon} \gamma^\nu u(p_1) \epsilon_\mu^2(q_2) \epsilon_\nu^1(q_1), \quad (1.17)$$

where, by momentum conservation, $q_1 = p_1 - p_2$ and $q_2 = p_2 - p_3$. Putting the spinor indices back, we get

$$\bar{u}(p_3) \gamma^\mu \frac{i(\not{p}_2 + m)}{p_2^2 - m^2 + i\epsilon} \gamma^\nu u(p_1) = \bar{u}_\alpha(p_3) \gamma_{\alpha\beta}^\mu \frac{i(\not{p}_2 + m)_{\beta\gamma}}{p_2^2 - m^2 + i\epsilon} \gamma_{\gamma\delta}^\nu u_\delta(p_1). \quad (1.18)$$

If we tie the ends of the diagram together, we would get a loop. Thus, we must replace the $u_\delta \bar{u}_\alpha$ by a propagator, which sums over all possible spins. We should also integrate over all momenta constrained by momentum conservation at each vertex. So the loop becomes

$$\begin{aligned}
i\mathcal{M} &= \begin{array}{c} \text{Diagram: A circular loop with two external wavy lines. The left wavy line is labeled } \epsilon_\nu^1 \text{ and the right wavy line is labeled } \epsilon_\mu^{2*}. \text{ The top arc of the loop is labeled } k \text{ and the bottom arc is labeled } k+q. \end{array} \\
&= -(-ie)^2 \int \frac{d^4k}{(2\pi)^4} e_\mu^{2*}(q) e_\nu^1(q) \left[\gamma_{\alpha\beta}^\mu \frac{i(\not{k} + \not{q} + m)_{\beta\gamma}}{(k+q)^2 - m^2 + i\epsilon} \gamma_{\gamma\delta}^\nu \frac{i(\not{k} + m)_{\delta\alpha}}{k^2 - m^2 + i\epsilon} \right] \\
&= e^2 \epsilon_\mu^{2*} \epsilon_\nu^1 \int \frac{d^4k}{(2\pi)^4} \text{Tr} \left[\gamma^\mu \frac{i(\not{k} + \not{q} + m)}{(k+q)^2 - m^2 + i\epsilon} \gamma^\nu \frac{i(\not{k} + m)}{(k^2 - m^2 + i\epsilon)} \right]. \quad (1.19)
\end{aligned}$$

A useful general rule is that the spinor matrices are always multiplied together in the direction opposite to the particle-flow arrow.

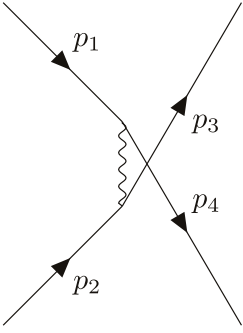
Signs

Now, the subtlety comes in. Recall that the spinors anticommute within a time-ordered product:

$$T\{\dots \psi(x) \psi(y) \dots\} = -T\{\dots \psi(y) \psi(x) \dots\}. \quad (1.20)$$

These minus signs would also appear in the Feynman rules. To see it, let us consider the Møller scattering ($e^- e^- \rightarrow e^- e^-$) at tree-level. Two possible Feynman diagrams contribute at tree-level, the t -channel diagram:

$$\begin{array}{c}
p_1 \quad p_3 \\
\downarrow \quad \uparrow \\
\text{Diagram: } t\text{-channel exchange} \\
\uparrow \quad \downarrow \\
p_2 \quad p_4
\end{array}
= \pm (-ie)^2 \bar{u}(p_3) \gamma^\mu u(p_1) \frac{-ig_{\mu\nu}}{(p_1 - p_3)^2} \bar{u}(p_4) \gamma^\nu u(p_2) \quad (1.21)$$



$$i\mathcal{M}_u = \pm (-ie)^2 \bar{u}(p_4) \gamma^\mu u(p_2) \frac{-ig_{\mu\nu}}{(p_1 - p_4)^2} \bar{u}(p_3) \gamma^\nu u(p_1) \quad (1.22)$$

What sign does these diagrams give?

The sum of amplitudes of these two diagrams is given by the perturbation expansion at order e^2 :

$$\begin{aligned} & (-ie)^2 \int d^4x \int d^4y \langle 0|T \left\{ \psi(x_1) \bar{\psi}(x_3) \psi(x_2) \bar{\psi}(x_4) \left(\bar{\psi}(x) A(x) \psi(x) \right) \left(\bar{\psi}(y) A(y) \psi(y) \right) \right\} |0\rangle \\ &= (-ie)^2 \gamma_{\beta_1\beta_2}^\mu \gamma_{\beta_3\beta_4}^\nu \int d^4x \int d^4y \langle 0|T \left\{ \psi_{\alpha_1}(x_1) \bar{\psi}_{\alpha_3}(x_3) \psi_{\alpha_2}(x_2) \bar{\psi}_{\alpha_4}(x_4) \bar{\psi}_{\beta_1}(x) \right. \\ &\quad \left. \times A^\mu(x) \psi_{\beta_2}(x) \bar{\psi}_{\beta_3}(y) A^\nu(y) \psi_{\beta_4}(y) \right\} |0\rangle. \end{aligned} \quad (1.23)$$

Recall Wick's theorem, which states that the vacuum matrix element of a time-ordered product is equal to all possible contractions in which all the fields are involved. For scalar field theory, a contraction is just a Feynman propagator:

$$\overline{\phi(x_1)\phi(x_2)} = \langle 0|T\{\phi(x_1)\phi(x_2)\}|0\rangle. \quad (1.24)$$

The same thing also applies to spinors. Thus, the vacuum matrix element of a time-ordered product is equal to all combinations of Feynman propagators in which all the spinors are involved. Since the Feynman propagator of spinors anticommutes, we must first put them in the correct order and then contract them.

For the above expansion, there are two possible contractions corresponding to the t - and u -channel. For the t -channel, the top electron line is created by $\bar{\psi}(x_3)$ annihilated by $\psi(x)$, created by $\bar{\psi}(x)$ annihilated by $\psi(x_1)$, and similarly for the bottom line. So we have

$$\begin{aligned} & (-ie)^2 \gamma_{\beta_1\beta_2}^\mu \gamma_{\beta_3\beta_4}^\nu \int d^4x \int d^4y \\ & \times \langle 0|T\{A^\mu(x) A^\nu(y) \psi_{\alpha_1}(x_1) \psi_{\beta_1}(x) \psi_{\beta_2}(x) \bar{\psi}_{\alpha_3}(x_3) \psi_{\alpha_2}(x_2) \bar{\psi}_{\beta_3}(y) \psi_{\beta_4}(y) \bar{\psi}_{\alpha_4}(x_4)\} |0\rangle. \end{aligned} \quad (1.25)$$

For u -channel, however, we need to swap $\bar{\psi}(x_3)$ and $\bar{\psi}(x_4)$, which gives an extra negative sign:

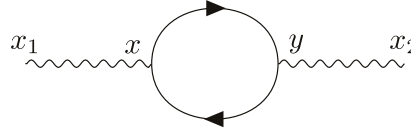
$$\begin{aligned} & - (-ie)^2 \gamma_{\beta_1\beta_2}^\mu \gamma_{\beta_3\beta_4}^\nu \int d^4x \int d^4y \\ & \times \langle 0|T\{A^\mu(x) A^\nu(y) \psi_{\alpha_1}(x_1) \psi_{\beta_1}(x) \psi_{\beta_2}(x) \bar{\psi}_{\alpha_4}(x_4) \psi_{\alpha_2}(x_2) \bar{\psi}_{\beta_3}(y) \psi_{\beta_4}(y) \bar{\psi}_{\alpha_3}(x_3)\} |0\rangle. \end{aligned} \quad (1.26)$$

Thus the matrix element of Møller scattering at tree-level has the form

$$\mathcal{M} = \mathcal{M}_t + \mathcal{M}_u = e^2 \left\{ \frac{[\bar{u}(p_3) \gamma^\mu u(p_1)][\bar{u}(p_4) \gamma^\mu u(p_2)]}{(p_1 - p_3)^2} - \frac{[\bar{u}(p_4) \gamma^\mu u(p_1)][\bar{u}(p_3) \gamma^\mu u(p_2)]}{(p_1 - p_4)^2} \right\}. \quad (1.27)$$

In the above, we have assumed the t -channel to be positive, but we could also have chosen the u -channel to be positive. However, the only physical quantity is $|\mathcal{M}|^2$. Thus, only the difference in sign would matter.

Now, consider a loop such as


(1.28)

which comes from the expansion

$$(-ie)^2 \gamma^\mu \gamma^\nu \gamma^\alpha \gamma^\beta \int d^4x \int d^4y \langle 0 | T \{ A^\mu(x_1) A^\nu(x_2) \bar{\psi}(x) A^\alpha(x) \psi(x) \bar{\psi}(y) A^\beta(y) \psi(y) \} | 0 \rangle. \quad (1.29)$$

To get the spinors into the order such that they are created then immediately destroyed, we need to anticommute $\psi(y)$ from the right to the left. That is, we use

$$\bar{\psi}_\alpha(x) \psi_\beta(x) \bar{\psi}_\gamma(y) \psi_\delta(y) = -\psi_\delta(y) \bar{\psi}_\alpha(x) \psi_\beta(x) \bar{\psi}_\gamma(y).$$

Thus, the Feynman rule for this fermion loop should be supplemented with an additional minus sign. As an exercise, you should check that adding more photons to a fermionic loop does not change this overall minus sign.

In summary, the Feynman rules for fermions must be supplemented by a factor of

- -1 for interchange of external identical fermions. Diagrams such as s - and t -channel exchanges, which would be present even for non-identical particles, do not get an extra minus sign. The -1 is a relative minus sign between two diagrams that are related by interchanging two external identical particles.
- -1 for each fermion loop.

γ -matrix Identities

Before doing any QED calculations, let us first derive some useful identities. We will often be dealing with traces of products of γ -matrices. These can often be simplified using the cyclic property of traces:

$$\text{Tr}[AB \cdots C] = \text{Tr}[B \cdots CA]. \quad (2.1)$$

Another useful thing to note is that

$$\gamma_5^2 = 1, \quad \gamma_5 \gamma_\mu = -\gamma_\mu \gamma_5. \quad (2.2)$$

First, compute

$$\text{Tr}[g^{\mu\nu}] = \text{Tr}[g^{\mu\nu} \mathbb{1}] + g^{\mu\nu} \text{Tr}[\mathbb{1}] = 4g^{\mu\nu}, \quad (2.3)$$

where we have used $\mathbb{1}$ to denote the identity matrix of the spinor indices.

Next, we evaluate

$$\text{Tr}[\gamma^\mu] = \text{Tr}[\gamma^5 \gamma^5 \gamma^\mu] = -\text{Tr}[\gamma^5 \gamma^\mu \gamma^5] = -\text{Tr}[\gamma^5 \gamma^5 \gamma^\mu] = -\text{Tr}[\gamma^\mu] \implies \text{Tr}[\gamma^\mu] = 0. \quad (2.4)$$

Then for the trace of a product of two γ -matrices, we find

$$\text{Tr}[\gamma^\mu \gamma^\nu] = \text{Tr}[2g^{\mu\nu} - \gamma^\nu \gamma^\mu] = 8g^{\mu\nu} - \text{Tr}[\gamma^\mu \gamma^\nu] \implies \text{Tr}[\gamma^\mu \gamma^\nu] = 4g^{\mu\nu}. \quad (2.5)$$

For three γ -matrices, we have

$$\text{Tr}[\gamma^\mu \gamma^\nu \gamma^\alpha] = \text{Tr}[\gamma^5 \gamma^5 \gamma^\mu \gamma^\nu \gamma^\alpha] = -\text{Tr}[\gamma^5 \gamma^\mu \gamma^\nu \gamma^\alpha \gamma^5] = -\text{Tr}[\gamma^\mu \gamma^\nu \gamma^\alpha] \implies \text{Tr}[\gamma^\mu \gamma^\nu \gamma^\alpha] = 0. \quad (2.6)$$

Since we can always do the same thing for traces of odd number of γ -matrices, they must be zero.

Now, for four γ -matrices, we find

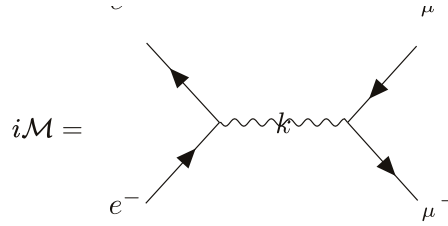
$$\begin{aligned} \text{Tr}[\gamma^\alpha \gamma^\mu \gamma^\beta \gamma^\nu] &= \text{Tr}[(2g^{\alpha\mu} - \gamma^\mu \gamma^\alpha) \gamma^\beta \gamma^\nu] \\ &= 2g^{\alpha\mu} \text{Tr}[\gamma^\beta \gamma^\nu] - \text{Tr}[\gamma^\mu \gamma^\alpha \gamma^\beta \gamma^\nu] \\ &= 8g^{\alpha\mu} g^{\beta\nu} - 2g^{\alpha\beta} \text{Tr}[\gamma^\mu \gamma^\nu] + \text{Tr}[\gamma^\mu \gamma^\beta \gamma^\alpha \gamma^\nu] \\ &= 8g^{\alpha\mu} g^{\beta\nu} - 8g^{\alpha\beta} g^{\mu\nu} + 2g^{\alpha\nu} \text{Tr}[\gamma^\mu \gamma^\beta] - \text{Tr}[\gamma^\mu \gamma^\beta \gamma^\nu \gamma^\alpha] \\ &= 8g^{\alpha\mu} g^{\beta\nu} - 8g^{\alpha\beta} g^{\mu\nu} + 8g^{\alpha\nu} g^{\beta\mu} - \text{Tr}[\gamma^\alpha \gamma^\mu \gamma^\beta \gamma^\nu] \\ &\implies \text{Tr}[\gamma^\alpha \gamma^\mu \gamma^\beta \gamma^\nu] = 4(g^{\alpha\mu} g^{\beta\nu} - g^{\alpha\beta} g^{\mu\nu} + g^{\alpha\nu} g^{\beta\mu}). \end{aligned} \quad (2.7)$$

Such method can be trivially applied to traces with more γ -matrices.

$$e^+ e^- \rightarrow \mu^+ \mu^-$$

As far as the QED is concerned, the muon, μ^- , is a particle identical to the electron, except heavier.

$e^+ e^- \rightarrow \mu^+ \mu^-$, is the simplest tree-level scattering process in QED. We will calculate at tree-level here and give the 1-loop correction later. The leading order contribution is



$$i\mathcal{M} = \quad (3.1)$$

$$= (-ie) \bar{v}_\alpha(p_2) \gamma_{\alpha\beta}^\mu u_\beta(p_1) \frac{-i \left[g_{\mu\nu} - (1 - \xi) \frac{k_\mu k_\nu}{k^2} \right]}{k^2} (-ie) \bar{u}_\delta(p_3) \gamma_{\delta\tau}^\nu v_\tau(p_4),$$

where $k^\mu = p_1^\mu + p_2^\mu = p_3^\mu + p_4^\mu$. The external spinors are forced to be on-shell, thus we can use the equations of motion $\not{p}_1 u(p_1) = m u(p_1)$ and $\bar{v}(p_2) \not{p}_2 = -m \bar{v}(p_2)$. Thus,

$$\begin{aligned} \bar{v}_\alpha(p_2) \gamma_{\alpha\beta}^\mu u_\beta(p_1) k_\mu &= \bar{v}(p_2) \not{p}_1 u(p_1) + \bar{v}(p_2) \not{p}_2 u(p_1) \\ &= \bar{v}(p_2) m u(p_1) - \bar{v}(p_2) m u(p_1) = 0. \end{aligned} \quad (3.2)$$

Similarly, for the other term contracted with k^ν . Thus, the $k^\mu k^\nu$ term vanishes, which is expected by gauge invariance. Hence, (3.1) becomes

$$\mathcal{M} = \frac{e^2}{s} [\bar{v}(p_2) \gamma^\mu u(p_1)] [\bar{u}(p_3) \gamma^\mu v(p_4)]. \quad (3.3)$$

Now, we need to find the conjugate amplitude to calculate $|\mathcal{M}|^2$. Using $\gamma^{0\dagger} = \gamma^0$ and $\gamma^{\mu\dagger} \gamma^0 = \gamma^0 \gamma^\mu$, we find

$$(\bar{\psi}_1 \gamma^\mu \psi_2)^\dagger = \psi_2^\dagger \gamma^{\mu\dagger} \gamma^0 \psi_1 = \psi_2^\dagger \gamma^0 \gamma^\mu \psi_1 = \bar{\psi}_2 \gamma^\mu \psi_1, \quad (3.4)$$

Hence, the conjugate of (3.3) is given by

$$\mathcal{M}^\dagger = \frac{e^2}{s} [\bar{v}(p_4) \gamma^\mu u(p_3)] [\bar{u}(p_1) \gamma^\mu v(p_2)], \quad (3.5)$$

and thus

$$|\mathcal{M}|^2 = \frac{e^4}{s^2} [\bar{v}(p_2) \gamma^\mu u(p_1)] [\bar{u}(p_3) \gamma^\mu v(p_4)] [\bar{v}(p_4) \gamma^\nu u(p_3)] [\bar{u}(p_1) \gamma^\nu v(p_2)]. \quad (3.6)$$

Unpolarized Scattering

The easiest thing to calculate from this is the cross section for scattering, assuming the spin is not measured. Then the final cross section should be the sum of all possible spin outcomes, which is

$$\begin{aligned} \sum_{s,s'} |\mathcal{M}|^2 &= \sum_{s,s'} \frac{e^4}{s^2} [\bar{v}^r(p_2) \gamma^\mu u^{r'}(p_1)] [\bar{u}^{r'}(p_1) \gamma^\nu v^r(p_2)] [\bar{u}^s(p_3) \gamma^\mu v^{s'}(p_4)] [\bar{v}^{s'}(p_4) \gamma^\nu u^s(p_3)] \\ &= \frac{e^4}{s^2} [\bar{v}^r(p_2) \gamma^\mu u^{r'}(p_1)] [\bar{u}^{r'}(p_1) \gamma^\nu v^r(p_2)] \sum_{s,s'} [\bar{u}_\alpha^s(p_3) \gamma_\alpha^\mu v_\beta^{s'}(p_4) \bar{v}_\gamma^{s'}(p_4) \gamma_\gamma^\nu u_\delta^s(p_3)] \\ &= \frac{e^4}{s^2} [\bar{v}^r(p_2) \gamma^\mu u^{r'}(p_1)] [\bar{u}^{r'}(p_1) \gamma^\nu v^r(p_2)] \sum_s [\bar{u}_\alpha^s(p_3) \gamma_\alpha^\mu (\not{\epsilon}_4 - m_\mu)_{\beta\gamma} \gamma_\gamma^\nu u_\delta^s(p_3)] \\ &= \frac{e^4}{s^2} [\bar{v}^r(p_2) \gamma^\mu u^{r'}(p_1)] [\bar{u}^{r'}(p_1) \gamma^\nu v^r(p_2)] [(\not{\epsilon}_3 + m_\mu)_{\delta\alpha} \gamma_\alpha^\mu (\not{\epsilon}_4 - m_\mu)_{\beta\gamma} \gamma_\gamma^\nu] \\ &= \frac{e^4}{s^2} [\bar{v}^r(p_2) \gamma^\mu u^{r'}(p_1)] [\bar{u}^{r'}(p_1) \gamma^\nu v^r(p_2)] \text{Tr}[(\not{\epsilon}_3 + m_\mu) \gamma^\mu (\not{\epsilon}_4 - m_\mu) \gamma^\nu], \end{aligned} \quad (3.7)$$

where we have used

$$\sum_s u_\alpha(p) \bar{u}_\beta(p) = \sum_s \bar{u}_\beta(p) u_\alpha(p) = (\not{\epsilon} + m)_{\alpha\beta}. \quad (3.8)$$

We shall also assume that we do not know the spins of the initial states. If we do the measurements for many times, we would get the average over each spin. Therefore, we shall take the average of the initial spins, which is

$$\begin{aligned} \frac{1}{4} \sum_{r,r'} \sum_{s,s'} |\mathcal{M}|^2 &= \frac{1}{4} \sum_{\text{spins}} |\mathcal{M}|^2 \\ &= \frac{e^4}{4s^2} \text{Tr}[(\not{\epsilon}_1 + m_e) \gamma^\nu (\not{\epsilon}_2 - m_e) \gamma^\mu] \text{Tr}[(\not{\epsilon}_3 + m_\mu) \gamma^\mu (\not{\epsilon}_4 - m_\mu) \gamma^\nu]. \end{aligned} \quad (3.9)$$

The trace can be simplified as

$$\begin{aligned} \text{Tr}[(\not{\epsilon}_3 + m_\mu) \gamma^\mu (\not{\epsilon}_4 - m_\mu) \gamma^\nu] &= p_3^\alpha p_4^\beta \text{Tr}[\gamma^\alpha \gamma^\mu \gamma^\beta \gamma^\nu] - m_\mu^2 \text{Tr}[\gamma^\mu \gamma^\nu] \\ &= 4p_3^\alpha p_4^\beta (g^{\alpha\mu} g^{\beta\nu} - g^{\alpha\beta} g^{\mu\nu} + g^{\alpha\nu} g^{\beta\mu}) - 4m_\mu^2 g^{\mu\nu} \\ &= 4p_3^\mu p_4^\nu - 4p_{34} g^{\mu\nu} + 4p_3^\nu p_4^\mu - 4m_\mu^2 g^{\mu\nu} \end{aligned} \quad (3.10)$$

with $p_{ij} \equiv p_i \cdot p_j$. Thus,

$$\begin{aligned} \frac{1}{4} \sum_{\text{spins}} |\mathcal{M}|^2 &= \frac{e^4}{4s^2} [4p_1^\mu p_2^\nu - 4p_{12} g^{\mu\nu} + 4p_1^\nu p_2^\mu - 4m_e^2 g^{\mu\nu}] [4p_3^\mu p_4^\nu - 4p_{34} g^{\mu\nu} + 4p_3^\nu p_4^\mu - 4m_\mu^2 g^{\mu\nu}] \\ &= \frac{e^4}{4s^2} [4p_1^\mu p_2^\nu + 4p_1^\nu p_2^\mu - 4(p_{12} + m_e^2) g^{\mu\nu}] [4p_3^\mu p_4^\nu + 4p_3^\nu p_4^\mu - 4(p_{34} + m_\mu^2) g^{\mu\nu}] \\ &= \frac{4e^4}{s^2} [2p_{13} p_{24} + 2p_{14} p_{23} + 4(p_{12} + m_e^2)(p_{34} + m_\mu^2) - 2(p_{12} + m_e^2)p_{34} - 2(p_{34} + m_\mu^2)p_{12}] \\ &= \frac{8e^4}{s^2} (p_{13} p_{24} + p_{14} p_{23} + p_{12} m_\mu^2 + p_{34} m_e^2 + 2m_\mu^2 m_e^2). \end{aligned} \quad (3.11)$$

This can be written in terms of the Mandelstam variables:

$$\begin{aligned} s &= (p_1 + p_2)^2 = (p_3 + p_4)^2 = 2m_e^2 + 2p_{12} = 2m_\mu^2 + 2p_{34}, \\ t &= (p_1 - p_3)^2 = (p_2 - p_4)^2 = m_e^2 + m_\mu^2 - 2p_{13} = m_e^2 + m_\mu^2 - 2p_{24}, \\ u &= (p_1 - p_4)^2 = (p_2 - p_3)^2 = m_e^2 + m_\mu^2 - 2p_{14} = m_e^2 + m_\mu^2 - 2p_{23}. \end{aligned} \quad (3.12)$$

Plugging these in, we have

$$\begin{aligned} \frac{1}{4} \sum_{\text{spins}} |\mathcal{M}|^2 &= \frac{8e^4}{s^2} \left[\left(\frac{m_e^2 + m_\mu^2 - t}{2} \right)^2 + \left(\frac{m_e^2 + m_\mu^2 - u}{2} \right)^2 \right. \\ &\quad \left. + \left(\frac{s - 2m_e^2}{2} \right) m_\mu^2 + \left(\frac{s - 2m_\mu^2}{2} \right) m_e^2 + 2m_\mu^2 m_e^2 \right] \\ &= \frac{2e^4}{s^2} \left[2m_e^4 + 2m_\mu^4 + 4m_e^2 m_\mu^2 - 2t(m_e^2 + m_\mu^2) - 2u(m_e^2 + m_\mu^2) \right. \\ &\quad \left. + t^2 + u^2 + 2s(m_e^2 + m_\mu^2) - 8m_e^2 m_\mu^2 + 8m_e^2 m_\mu^2 \right] \\ &= \frac{2e^4}{s^2} \left[t^2 + u^2 + 2(s - t - u)(m_e^2 + m_\mu^2) + 2m_e^4 + 2m_\mu^4 \right]. \end{aligned} \quad (3.13)$$

Also, note

$$\begin{aligned} s + u + t &= (2m_e^2 + 2p_{12}) + (m_e^2 + m_\mu^2 - 2p_{13}) + (m_e^2 + m_\mu^2 - 2p_{14}) \\ &= 4m_e^2 + 2m_\mu^2 + 2p_1 \cdot (p_2 - p_3 - p_4) \\ &= 4m_e^2 + 2m_\mu^2 - 2p_1^2 \\ &= 2(m_e^2 + m_\mu^2). \end{aligned} \quad (3.14)$$

Thus, we have

$$\frac{1}{4} \sum_{\text{spins}} |\mathcal{M}|^2 = \frac{2e^4}{s^2} \left[t^2 + u^2 + 4s(m_e^2 + m_\mu^2) - 2(m_e^2 + m_\mu^2)^2 \right]. \quad (3.15)$$

Differential Cross Section

Now, let us compute the cross section in the centre of mass frame. Since the initial and final masses are different, the cross section is given by

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 E_{\text{CM}}^2} \frac{|\vec{p}_f|}{|\vec{p}_i|} |\mathcal{M}|^2. \quad (3.16)$$

In the centre of mass frame, let us define the energy of all the particles to be E (by energy conservation), and define the momenta of the incoming particles to be $\vec{p}$, the momenta of the outgoing particles to be $\vec{q}$. So that

$$\begin{aligned} p_1^\mu &= (E, \vec{p}), \\ p_2^\mu &= (E, -\vec{p}), \\ p_3^\mu &= (E, \vec{q}), \\ p_4^\mu &= (E, -\vec{q}). \end{aligned} \quad (3.17)$$

The Mandelstam variables become,

$$\begin{aligned}
s &= (p_1 + p_2)^2 = 4E^2 = E_{\text{CM}}^2, \\
t &= (p_1 - p_3)^2 = -(\vec{p} - \vec{q})^2 = -\vec{p}^2 - \vec{q}^2 + 2\vec{p} \cdot \vec{q} = -2E^2 + m_e^2 + m_\mu^2 + 2\vec{p} \cdot \vec{q}, \\
u &= (p_1 - p_4)^2 = -(\vec{p} + \vec{q})^2 = -\vec{p}^2 - \vec{q}^2 - 2\vec{p} \cdot \vec{q} = -2E^2 + m_e^2 + m_\mu^2 - 2\vec{p} \cdot \vec{q}.
\end{aligned} \tag{3.18}$$

Thus, (3.15) becomes

$$\begin{aligned}
\frac{1}{4} \sum_{\text{spins}} |\mathcal{M}|^2 &= \frac{2e^4}{s^2} \left[(2E^2 - m_e^2 - m_\mu^2 - 2\vec{p} \cdot \vec{q})^2 + (2E^2 - m_e^2 - m_\mu^2 + 2\vec{p} \cdot \vec{q})^2 \right. \\
&\quad \left. + 16E^2(m_e^2 + m_\mu^2) - 2(m_e^2 + m_\mu^2)^2 \right] \\
&= \frac{2e^4}{s^2} \left[8E^4 + 8E^2(m_e^2 + m_\mu^2) + 8(\vec{p} \cdot \vec{q})^2 \right].
\end{aligned} \tag{3.19}$$

We define the angle between $\vec{p}$ and $\vec{q}$ to be θ , so that

$$\vec{p} \cdot \vec{q} = |\vec{p}| |\vec{q}| \cos \theta. \tag{3.20}$$

The cross section then becomes

$$\begin{aligned}
\frac{d\sigma}{d\Omega} &= \frac{e^4}{4\pi^2(4E^2)^3} \frac{|\vec{p}|}{|\vec{q}|} \left[E^4 + E^2(m_e^2 + m_\mu^2) + |\vec{p}|^2 |\vec{q}|^2 \cos^2 \theta \right] \\
&= \frac{\alpha^4}{16E^6} \frac{\sqrt{E^2 - m_\mu^2}}{\sqrt{E^2 - m_e^2}} \left[E^4 + E^2(m_e^2 + m_\mu^2) + (E^2 - m_e^2)(E^2 - m_\mu^2) \cos^2 \theta \right],
\end{aligned} \tag{3.21}$$

where $\alpha = \frac{e^2}{4\pi}$ is the fine structure constant.

For simplicity, take $m_e = 0$, we find

$$\begin{aligned}
\frac{d\sigma}{d\Omega} &= \frac{\alpha^4}{16E^6} \sqrt{1 - \frac{m_\mu^2}{E^2}} \left[E^4 + E^2 m_\mu + E^2(E^2 - m_\mu^2) \cos^2 \theta \right] \\
&= \frac{\alpha^4}{4E_{\text{CM}}^2} \sqrt{1 - \frac{m_\mu^2}{E^2}} \left[\left(1 + \frac{m_\mu^2}{E^2} \right) + \left(1 - \frac{m_\mu^2}{E^2} \right) \cos^2 \theta \right].
\end{aligned} \tag{3.22}$$

Under the ultra-high-energy limit we can take $m_\mu = 0$, which gives

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^4}{4E_{\text{CM}}^2} (1 + \cos^2 \theta). \tag{3.23}$$

Rutherford Scattering $e^- p^+ \rightarrow e^- p^+$

The classical Rutherford scattering formula is

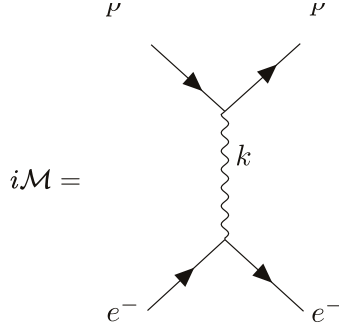
$$\frac{d\sigma}{d\Omega} = \frac{m_e^2 e^4}{4p^4 \sin^4 \frac{\theta}{2}}, \tag{4.1}$$

where $p = |\vec{p}_i| = |\vec{p}_f|$ is the magnitude of the incoming electron momentum, which is the same as the magnitude of the outgoing electron.

Here, we will do the calculation in QED, while neglecting any internal structure of the proton, treating it as a point-like particle.

QED Amplitude

As far as QED is concerned, a proton and a muon are the same thing, up to a sign of the charge which gets squared anyway. Since the proton and the electron are distinguishable, the only contribution to the amplitude at tree-level is the t -channel:



$$i\mathcal{M} = (-ie)\bar{u}(p_3)\gamma^\mu u(p_1) \frac{-i \left[g_{\mu\nu} - (1 - \xi) \frac{k_\mu k_\nu}{k^2} \right]}{k^2} (ie)\bar{u}(p_4)\gamma^\nu u(p_2). \quad (4.2)$$

This is almost identical to what we had before, and thus we can read off

$$|\mathcal{M}|^2 = \frac{e^2}{t} [\bar{u}(p_3)\gamma^\mu u(p_1)][\bar{u}(p_1)\gamma^\nu u(p_3)][\bar{u}(p_4)\gamma^\mu u(p_2)][\bar{u}(p_2)\gamma^\nu u(p_4)]. \quad (4.3)$$

Assuming we do not have any knowledge of the initial spins and do not measure the final spins, we have

$$\frac{1}{4} \sum_{\text{spins}} |\mathcal{M}|^2 = \frac{e^4}{4t^2} \text{Tr}[(\not{p}_3 + m_e)\gamma^\mu(\not{p}_1 + m_e)\gamma^\nu] \text{Tr}[(\not{p}_4 + m_p)\gamma^\mu(\not{p}_2 + m_p)\gamma^\nu]. \quad (4.4)$$

Comparing this with (3.9), we find they are identical up to the masses after the replacement

$$(p_1, p_2, p_3, p_4) \rightarrow (p_1, -p_3, p_4, -p_2). \quad (4.5)$$

These changes send

$$s = (p_1 + p_2)^2 \rightarrow (p_1 - p_3)^2 = t, \quad (4.6)$$

$$t = (p_1 - p_3)^2 \rightarrow (p_1 - p_4)^2 = u, \quad (4.7)$$

$$u = (p_1 - p_4)^2 \rightarrow (p_1 + p_2)^2 = s. \quad (4.8)$$

Thus, we find

$$\frac{1}{4} \sum_{\text{spins}} |\mathcal{M}|^2 = \frac{2e^4}{t^2} \left[u^2 + s^2 + 4t(m_e^2 + m_p^2) - 2(m_e^2 + m_p^2)^2 \right]. \quad (4.9)$$

Cross Section

Now, let us take the $m_p \gg m_e$ limit to get the relativistic corrections to Rutherford's formula. Thus, the momenta are

$$p_1^\mu = (E, \vec{p}_i), \quad p_2^\mu = (m_p, 0), \quad p_3^\mu = (E, \vec{p}_f), \quad p_4^\mu = (m_p, 0). \quad (4.10)$$

The scattering angle, θ , is defined as

$$\vec{p}_i \cdot \vec{p}_f = p^2 \cos \theta, \quad (4.11)$$

where $p = |\vec{p}_i| = |\vec{p}_f|$.

Using

$$v = \frac{p}{E} = \sqrt{1 - \frac{m_e^2}{E^2}}, \quad (4.12)$$

the above becomes

$$\vec{p}_i \cdot \vec{p}_f = p^2 \cos \theta = v^2 E^2 \cos \theta. \quad (4.13)$$

In the limit $m_p \gg m_e$ is the electron's relativistic velocity, the relevant momentum products are:

$$p_{13} = E^2(1 - v^2 \cos \theta), \quad (4.14)$$

$$p_{12} = p_{23} = p_{34} = p_{14} = Em_p, \quad (4.15)$$

$$p_{24} = m_p^2, \quad (4.16)$$

where $p_{ij} \equiv p_i^\mu p_{j\mu}$ and

$$t = (p_1 - p_3)^2 = -(\vec{p}_i - \vec{p}_f)^2 = -2p^2(1 - \cos \theta), \quad (4.17)$$

so that

$$\begin{aligned} \frac{1}{4} \sum_{\text{spins}} |\mathcal{M}|^2 &= \frac{8e^4}{t^2} [p_{14}p_{23} + p_{12}p_{34} - m_p^2 p_{13} - m_e^2 p_{24} + 2m_e^2 m_p^2] \\ &= \frac{8e^4}{4v^4 E^4 (1 - \cos \theta)^2} [E^2 m_p^2 + E^2 m_p^2 v^2 \cos \theta + m_e^2 m_p^2] \\ &= \frac{2e^4 m_p^2}{v^4 E^2 (1 - \cos \theta)^2} [2 - v^2 (1 - \cos \theta)] \\ &= \frac{e^4 m_p^2}{v^4 E^2 \sin^4 \frac{\theta}{2}} \left[1 - v^2 \sin^2 \frac{\theta}{2} \right]. \end{aligned} \quad (4.18)$$

Thus,

$$\frac{d\sigma}{d\Omega} = \frac{e^4}{64\pi^2 v^2 p^2 \sin^4 \frac{\theta}{2}} \left(1 - v^2 \sin^2 \frac{\theta}{2} \right), \quad E \ll m_p. \quad (4.19)$$

This is known as the Mott formula. Note that the limit $m_p \rightarrow \infty$ exists: there is no dependence on the proton mass. For slow velocities we can use $v \ll 1$ and $p \ll E \sim m_e$ so $v \sim \frac{p}{m_e}$. Thus,

$$\frac{d\sigma}{d\Omega} = \frac{e^4 m_e^2}{64\pi^2 p^4 \sin^4 \frac{\theta}{2}}, \quad v \ll 1 \text{ and } E \ll m_p. \quad (4.20)$$

which is the Rutherford formula. In particular, note that the normalization factors, m_e^2 , worked out correctly.

In the very high energy limit, $E \gg m_e$, one can no longer assume that the final state proton is also at rest. However, one can now neglect the electron mass, so that $v = 1$. Then the momenta are

$$p_1^\mu = (E, \vec{p}_i), \quad p_2^\mu = (m_p, 0), \quad p_3^\mu = (E', \vec{p}_f), \quad p_4^\mu = p_1^\mu + p_2^\mu - p_3^\mu \quad (4.21)$$

with $|\vec{p}_i| = E$ and $|\vec{p}_f| = E'$. For $m_e = 0$ in the proton rest frame, following the same steps as above, we find the cross section:

$$\frac{d\sigma}{d\Omega} = \frac{e^4}{64\pi^2 E^2 \sin^4 \frac{\theta}{2}} \frac{E'}{E} \left(\cos^2 \frac{\theta}{2} + \frac{E - E'}{m_p} \sin^2 \frac{\theta}{2} \right), \quad m_e \ll E. \quad (4.22)$$

where E' is the final state electron's energy. As $m_p \rightarrow \infty$, $E \rightarrow E'$ and this reduces to the $v \rightarrow 1$ limit of the Mott formula.

These formulas describe the scattering of point-like particles by other point-like particles. Note that the final forms in which we have written these cross sections depend only on properties of the initial and final electrons. Thus, they are suited to experimental situations in which electrons are scattered by hydrogen gas and the final state proton momenta are not measured. Such experiments were carried out in the 1950s, and deviations of the measured cross section led to the conclusion that the proton must have substructure. More shockingly, at very high energy, this point-like scattering form was once again observed, indicating the presence of point-like constituents within the proton, now known as quarks.